$\mathcal{K}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{K}_{0^+}^{\#1}\dagger$	$\frac{1}{2}(-Y_1 k^2 - 3\,{}^{(2)}\kappa_3)$	0
$\mathcal{K}_{0^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	$Y_2 k^2$

$\mathcal{K}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^-}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^-}^{\#2}{}_{\alpha}$
$\mathcal{K}_{1^+}^{\#1}\dagger^{\alpha\beta}$	0	0	0	0
$\mathcal{K}_{1^+}^{\#2}\dagger^{\alpha\beta}$	0	0	0	0
$\mathcal{K}_{1^-}^{\#1}\dagger^{\alpha}$	0	0	$\frac{1}{2}(Y_3 k^2 - 2\,{}^{(2)}\kappa_3)$	$\frac{-Y_3 k^2 + 2\,{}^{(2)}\kappa_3}{2\sqrt{2}}$
$\mathcal{K}_{1^-}^{\#2}\dagger^{\alpha}$	0	0	$\frac{-Y_3 k^2 + 2\,{}^{(2)}\kappa_3}{2\sqrt{2}}$	$\frac{1}{4}(Y_3 k^2 - 2\,{}^{(2)}\kappa_3)$

$\mathcal{K}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{K}_{2^+}^{\#1}\dagger^{\alpha\beta}$	$\frac{Y_4 k^2}{2}$	0
$\mathcal{K}_{2^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	$\frac{Y_4 k^2}{2}$

$\mathcal{J}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{J}_{0^+}^{\#1}\dagger$	$\frac{1}{\text{Det}(0^+)}$	0
$\mathcal{J}_{0^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	$\frac{1}{k^2({}^{(4)}\kappa_{15} - {}^{(4)}\kappa_{16})}$

$\mathcal{J}$	$\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{J}_{1^-}^{\#1}{}_{\alpha}$	$\mathcal{J}_{1^-}^{\#2}{}_{\alpha}$
$\mathcal{J}_{1^+}^{\#1}\dagger^{\alpha\beta}$	0	0	0	0
$\mathcal{J}_{1^+}^{\#2}\dagger^{\alpha\beta}$	0	0	0	0
$\mathcal{J}_{1^-}^{\#1}\dagger^{\alpha}$	0	0	$\frac{2}{3\,\text{Det}(1^-)}$	$-\frac{\sqrt{2}}{3\,\text{Det}(1^-)}$
$\mathcal{J}_{1^-}^{\#2}\dagger^{\alpha}$	0	0	$-\frac{\sqrt{2}}{3\,\text{Det}(1^-)}$	$\frac{1}{3\,\text{Det}(1^-)}$

$\mathcal{J}$	$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{2^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{J}_{2^+}^{\#1}\dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0
$\mathcal{J}_{2^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(2^-)}$

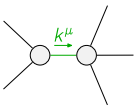
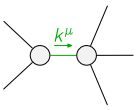
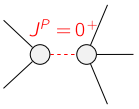
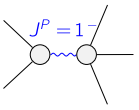
Abbreviations used in matrices

$$Y_1 == 3\,{}^{(4)}\kappa_1 - 2\,{}^{(4)}\kappa_{15} - {}^{(4)}\kappa_{16} + 3\,{}^{(4)}\kappa_2 \;\&\&\; Y_2 == {}^{(4)}\kappa_{15} - {}^{(4)}\kappa_{16} \;\&\&\; Y_3 == 2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16} - 2\,{}^{(4)}\kappa_2 \;\&\&$$
$$Y_4 == 2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16} \;\&\&\; \text{Det}(0^+) == \frac{1}{2}(k^2(-3\,{}^{(4)}\kappa_1 + 2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16} - 3\,{}^{(4)}\kappa_2) - 3\,{}^{(2)}\kappa_3) \;\&\&$$
$$\text{Det}(1^-) == \frac{3}{4}(k^2(2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16} - 2\,{}^{(4)}\kappa_2) - 2\,{}^{(2)}\kappa_3) \;\&\&\; \text{Det}(2^+) == \frac{1}{2}k^2(2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16})$$

Lagrangian

$$\begin{aligned}&{}^{(2)}\kappa_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi} + {}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - \frac{1}{2}\,{}^{(4)}\kappa_{16}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + \\ &2\,{}^{(4)}\kappa_{16}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - {}^{(4)}\kappa_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + \frac{1}{2}\,{}^{(4)}\kappa_{16}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - 2\,{}^{(4)}\kappa_{15}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \\ &{}^{(4)}\kappa_{16}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - {}^{(4)}\kappa_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + {}^{(4)}\kappa_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} + {}^{(4)}\kappa_{16}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{1^-}^{\#1\,\alpha} + 2\,\mathcal{J}_{1^-}^{\#2\,\alpha} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{1^+}^{\#2\,\alpha\beta} == 0$	3	$\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}$
$\mathcal{J}_{1^+}^{\#1\,\alpha\beta} == 0$	3	$\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$
Total # constraint(s):	9	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16}}$
	2	0	$-\frac{1}{2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16}}$
	1	$\frac{3\,{}^{(2)}\kappa_3}{-3\,{}^{(4)}\kappa_1 + 2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16} - 3\,{}^{(4)}\kappa_2}$	$-\frac{2}{3\,{}^{(4)}\kappa_1 - 2\,{}^{(4)}\kappa_{15} - {}^{(4)}\kappa_{16} + 3\,{}^{(4)}\kappa_2}$
	3	$\frac{2\,{}^{(2)}\kappa_3}{2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16} - 2\,{}^{(4)}\kappa_2}$	$-\frac{4}{3(2\,{}^{(4)}\kappa_{15} + {}^{(4)}\kappa_{16} - 2\,{}^{(4)}\kappa_2)}$
Resolved unitarity condition(s):	(Demonstrably impossible)		